Bailin Liang

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EDUCATION

Jilin University, Changchun, China (Project 985 University) **Sept 2024–Jun 2027 (Expected)**
M.S. in Automotive Engineering (**Exam-Free Admission**), College of Automotive Engineering Supervisor: Prof. Junnian Wang
Research Direction: vehicle motion control, corner-module vehicle, nonlinear dynamics and model-/learning-based control
Awards include **WEICHAI POWER Social Scholarship (Top 1%)**, **Outstanding Student Scholarship** and **Academic Scholarship**.

Jilin University, Changchun, China **Sept 2020–Jun 2024**
B.Eng. in Automotive Engineering, College of Automotive Engineering GPA: 87.89/100 | A-
Awards include **Second Class Scholarship** and **Outstanding Student of the College**.

PUBLICATIONS & PATENTS

Papers

[First Author] Generalized Fault-Tolerant Control for Multi-Wheel Corner-Module Vehicles via Reconfigurable MPC.
ISA Transactions (JCR **Q1**; IF=6.6), under review.
This work proposes a generalized fault-tolerant control framework for multi-wheel corner-module vehicles, enabling automated transfer across arbitrary multi-axle chassis configurations. By integrating automatic topology identification, health-aware MPC reconfiguration, and physical latent-space optimization, the framework maintains vehicle stability and tracking accuracy under diverse fault scenarios. Hardware-in-the-loop experiments achieve real-time execution within 12.36 ms, validating its cross-configuration adaptability and practical deployability.

[First Author] Nonlinear Shimmy Dynamics Analysis of Highly Integrated In-Wheel Motor Driving Unit with Independent Kingpin Steering for Electric Vehicles. *Nonlinear Dynamics* (JCR **Q1**; IF=5.7), under review.
This work proposes a nonlinear shimmy dynamics analysis framework for highly integrated in-wheel motor driving units (HIDUs), combines complex variable averaging (CX-A) and eigenspace-projected shooting methods for bifurcation analysis, develops an FPSA method for efficient parameter evaluation, and reveals parameter-dependent shimmy suppression mechanisms through energy-flow analysis.

[Contributor] A Review of Configuration Classification, Application Control and Future Trends of Highly Integrated Driving Corner Modules for Electric Vehicles. *Automotive Innovation* (JCR **Q1**; IF=7.6), under review.

Patents

As the **first student inventor**, I have authored **10 granted or published invention patents**, including **one granted U.S. patent**, and participated in **18 granted or published invention patents** in total.

- Kingpin Steering and Omni-Directional Steering Dual-Mode Power-Multiplexing Electric Wheel Angle Module. (**US20260200519A1**, granted Jul 2026; CN120246072A, granted Aug 2025.)
- Distributed Supercapacitor Energy Storage System Based on Electric Wheel Corner Modules. (CN120287823A, granted Oct 2025.)
- Land-Air Power-Multiplexing Flying Car Using Electric Wheel Corner Modules. (CN121492543A, published Feb 2026.)
- A Control Method for a Power-Multiplexed Integrated Drive-Braking Electric Wheel System. (CN120287825A, published Jul 2025.)
- Multi-Link-Braking Integrated Drive-Braking Electric Wheel System. (CN120481601A, published Aug 2025.)
- Power-Multiplexed Integrated Drive-Braking Electric Wheel System. (CN120287824A, published Jul 2025.)
- Highly Integrated Driving Unit with Electronically Adjustable Kingpin Caster Angle. (CN121492543A, published Jul 2025.)
- Electronic Wedge Braking System with a Sprung Actuator. (CN120288017A, published Jul 2025.)
- Multi-Mode Redundant Integrated Drive-Braking Electric Wheel System. (CN120287823A, published Jul 2025.)

RESEARCH PROJECTS

Key Technologies for Highly Integrated In-Wheel-Motor Driving Units Changchun, China *National Key R&D Program of China, Grant No. 2024YFB2505100* *Apr. 2025 – Present*

- Served as a main contributor to the sub-project **Motion Characteristics Analysis and Configuration Design of Highly Integrated In-Wheel-Motor Driving Units**, developing high-fidelity MATLAB/Simulink mathematical models and Adams Car multibody dynamics models for shimmy mechanism analysis and parameter sensitivity studies. Proposed a nonlinear shimmy dynamics analysis workflow for evaluating HIDU shimmy performance, which has been implemented as a preliminary analysis software tool.

Nonlinear Shimmy Dynamics Analysis and Learning-Enhanced Control for Shimmy Suppression in Multi-Wheel Corner-Module Vehicles *Ongoing Research*

- **Generalized Shimmy Dynamics Modeling:** Developed a generalized nonlinear shimmy model for multi-wheel corner-module vehicles, capturing the inertial coupling among wheel-shimmy motion, suspension roll, and wheel rotational oscillations, together with lateral tire relaxation, steering backlash, and dry friction. Its modular architecture enables rapid adaptation across different multi-wheel corner-module chassis configurations.
- **Nonlinear Shimmy Dynamics Analysis:** Investigating the nonlinear shimmy dynamics of multi-wheel corner-module vehicles, with emphasis on how steering backlash, dry friction, and dynamic coupling across multiple subsystems reshape stability boundaries and give rise to complex shimmy responses characterized by periodicity, quasiperiodicity, slow amplitude modulation, multistability, and nonzero-mean oscillations.
- **Shimmy-Aware Neural-Residual DiffTune-MPC:** Developing a hybrid prediction model that augments a physics-guided nominal model with a neural residual to capture unmodeled nonlinear dynamics and reduce prediction mismatch. A shimmy-aware DiffTune-MPC scheme tunes the inner MPC cost weights against a closed-loop outer objective that minimizes band-limited shimmy energy and promotes oscillation-envelope contraction, enabling coordinated drive–brake–steer control for shimmy suppression.

Wheel-End Energy Autonomy for Distributed-Drive Electric Vehicles *Jan. 2024 – Oct. 2024*

- Developed a distributed wheel-end supercapacitor–battery architecture for distributed-drive electric vehicles, reducing energy-transfer losses and battery stress through localized energy management. Completed topology selection and parameter matching, ADVISOR/Simulink modeling, power-split control, and regenerative-braking optimization. Further integrated the architecture into corner modules with onboard energy storage, enabling local energy recovery, storage, and reuse while providing wheel-end energy autonomy and energy redundancy.

INTERNSHIP

XPENG Motors, XNGP Team Guangzhou, China
Autonomous Driving R&D Intern *Jan 2024–May 2024*

- Studied the XNet-XBrain-XPlanner modular end-to-end autonomous driving framework adopted in XPENG's XNGP system.
- Participated in the large-scale data closed loop for urban road tests, takeover events, corner cases, and OTA iterations, focusing on failure-scenario identification, abnormal takeover analysis, and root-cause investigation.

Gspeed Formula Racing Team of Jilin University Changchun, China
Member of Suspension Group *Dec 2020–Dec 2023*

- Designed and modeled critical suspension components using CATIA, produced engineering drawings in AutoCAD, and built force-analysis tables for corner-wheel components based on hardpoints and vehicle performance targets.
- Participated in the full cycle of suspension system assembly, maintenance, and tuning.

SKILLS

Application & Coding: MATLAB/Simulink, Python, C/C++, CATIA, AutoCAD. **English:** IELTS 6, CET-6. **Cantonese/Mandarin**: Native.